Comprehensive Technical LLD Specification: PostGIS Schemas, State Machines, API Contracts & Algorithmic Engines

Governing Authority: Ministry of Cooperation, Government of India • *Theme: Sahakar Se Samriddhi*
Architecture: Asynchronous Micro-modular (FastAPI + PostGIS + Leaflet + Gemini AI) • **Version:** 1.0.0 (SIH 2026 Production Specification)

Low-Level Technical Blueprint: This specification provides the complete engineering implementation model for the Seva Setu platform. It details the object-oriented domain models, the production PostgreSQL 16 + PostGIS spatial schema, the 10-state booking lifecycle state machine with en-route fallback, sequence diagrams for on-site material procurement and emergency SOS, the 4-layer decoupled pricing engine, and concurrency control using Redis Redlock and optimistic locking.

1. System Architecture & Modular Topology

Seva Setu is decomposed into independent functional micro-services coordinated through an asynchronous API Gateway. The architecture eliminates single points of failure and isolates high-frequency spatial GPS updates from the financial settlement ledger.

Layer / Service	Underlying Tech Stack	Core Responsibilities & State Boundaries
Presentation Layer	React 18 / Next.js 14, Tailwind CSS, Leaflet.js, Web Speech API	Citizen PWA, Worker Radar HUD (one-handed touch targets), Ministry Cockpit dashboard.
API Gateway & Auth	FastAPI, Python 3.11+, PyJWT, Pydantic v2	Role-Based Access Control (RBAC), rate-limiting, request validation, CORS, token verification.
Spatial Dispatch Engine	PostgreSQL 16, PostGIS, GeoPy, Redis 7 (GEORADIUS)	Sub-second proximity matching, radial worker discovery, en-route hurdle fallback routing.
Pricing & Subsidy Engine	Python Async Service, Redis Cache	4-Layer decoupled fare calculation, transparent breakdown generation, dynamic NGO quota check.
AI Credential Parser	Google GenAI SDK (Gemini 1.5 Flash)	Multimodal document OCR of ITI, NSDC, and e-Shram certificates; confidence matching.
Safety & SOS Service	FastAPI, Telephony Bridge, Police 112 API	Instant encrypted telemetry packet dispatch (Aadhaar, live GPS, vehicle registration).

2. Domain Entities & UML Class Structure

The object model establishes clear ownership and audit boundaries. Key domain models enforce bilateral accountability:

- **User Class Hierarchy:** Base User class splits into Citizen (tracking Darpan NGO verification, subsidy quota hours, and bilateral strike counts) and CooperativeWorker (tracking PACS federation affiliation, 14-day rolling rejection rates, prepaid float ledger, and emergency certification badges).
- **Booking Aggregate:** Represents the transaction core, holding pickup spatial coordinates, hashed OTP verification tokens (start_otp_hash, end_otp_hash), pre- and post-repair proof URLs, and the full rejection escalation history.
- **TransparentInvoice Model:** Explicitly decouples Base Labor (100% worker passthrough), Toolkit Fee, Transit Tariff, Emergency Surcharge, Platform Tech Fee (3.5%), and Cooperative Welfare Cess (1.5%).
- **MaterialProcurementRequest:** Models the on-site spare part approval pipeline with customer pre-authorization flags, damaged-part photo evidence, and wholesale merchant B2B margin tracking.

3. Production Database Schema (PostgreSQL 16 + PostGIS DDL)

Below is the core SQL schema defining spatial indexes and integrity constraints:

```
-- 1. Enable Spatial and UUID Extensions
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";
CREATE EXTENSION IF NOT EXISTS postgis;

-- 2. Cooperative Federations (PACS & State Guilds)
CREATE TABLE cooperative_federations (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    registration_code VARCHAR(64) UNIQUE NOT NULL,
    name VARCHAR(255) NOT NULL,
    district VARCHAR(100) NOT NULL, state VARCHAR(100) NOT NULL,
    welfare_pool_balance NUMERIC(12,2) DEFAULT 0.00
);

-- 3. Cooperative Workers with PostGIS Geometry
CREATE TABLE cooperative_workers (
    id UUID PRIMARY KEY REFERENCES users(id) ON DELETE CASCADE,
    coop_federation_id UUID NOT NULL REFERENCES cooperative_federations(id),
    primary_trade VARCHAR(64) NOT NULL,
    is_verified BOOLEAN DEFAULT FALSE, is_available BOOLEAN DEFAULT FALSE,
    rating NUMERIC(3,2) DEFAULT 5.00, rejection_rate_14d NUMERIC(4,3) DEFAULT 0.000,
    prepaid_float_balance NUMERIC(10,2) DEFAULT 500.00,
    current_location GEOMETRY(Point, 4326)
);
CREATE INDEX idx_workers_spatial ON cooperative_workers USING GIST (current_location);
CREATE INDEX idx_workers_trade_avail ON cooperative_workers (primary_trade, is_available, is_verified);

-- 4. Bookings Table with Dual OTP & State Constraints
CREATE TABLE bookings (
    id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
    booking_reference VARCHAR(32) UNIQUE NOT NULL,
    citizen_id UUID NOT NULL REFERENCES citizens(id),
    assigned_worker_id UUID REFERENCES cooperative_workers(id),
    service_code VARCHAR(64) NOT NULL, sub_service_name VARCHAR(128) NOT NULL,
    urgency_tier VARCHAR(32) NOT NULL CHECK (urgency_tier IN ('P1_CRITICAL','P2_HIGH','P3_STANDARD','P4_ROUTINE')),
    status VARCHAR(32) NOT NULL DEFAULT 'DRAFT',
    pickup_coords GEOMETRY(Point, 4326) NOT NULL,
    start_otp_hash CHAR(64) NOT NULL, end_otp_hash CHAR(64) NOT NULL,
    pre_repair_photo_url TEXT, post_repair_photo_url TEXT,
    payment_mode VARCHAR(20) NOT NULL CHECK (payment_mode IN ('CASH_ON_DELIVERY','UPI_ESCROW')),
    rejection_history UUID[] DEFAULT '{}'
);
CREATE INDEX idx_bookings_status ON bookings (status);
```

4. Finite State Machines (FSM): 10-State Booking Lifecycle

The booking state machine guarantees that requests never enter an unmonitored deadlock. Every transition is governed by explicit timeouts, cryptographic handshakes, or en-route hurdle triggers.

Current State	Trigger / Event	Next State	System Action & Operational Fallback
DRAFT	Order submitted & invoice approved	BROADCASTING	Spatial query identifies top candidate; 30s acceptance timer initiated.
BROADCASTING	Worker declines or 30s timeout expires	REROUTING_FALLBACK	Logs worker rejection; auto-reassigns to next nearest qualified technician in zone.
BROADCASTING	Worker accepts offer	ACCEPTED	Redis distributed lock acquired; worker profile & vehicle details sent to citizen.
ACCEPTED	Worker starts travel	EN_ROUTE	In-app navigation active; Google Maps deep-link enabled; live coordinates stream.
EN_ROUTE	Worker reports vehicle breakdown / traffic hurdle	HURDLE_REROUTING	Mandatory reason logged; booking instantly re-escalates to next nearest guild peer.
EN_ROUTE	Worker reaches geofence (within 100m)	ARRIVED	App prompts citizen to provide START_OTP; prompts worker to input code.
ARRIVED	START_OTP verified	IN_PROGRESS	Service timer starts; pre-repair photo logged; doorstep session authenticated.
IN_PROGRESS	Worker requests spare parts	MATERIAL_APPROVAL	Photo + cost prompt sent to citizen; upon approval, procurement pass QR issued.
IN_PROGRESS	Work executed; photos captured	PENDING_PROOF	Dual completion photos uploaded (citizen + worker); END_OTP requested.
PENDING_PROOF	END_OTP validated	COMPLETED	Automated bank split: 100% labor to worker, 1.5% to PACS fund, 3.5% tech fee.

5. Critical Architectural Sequences & Handshakes

A. The En-Route Hurdle & Zero-Downtime Fallback Sequence

1. **Hurdle Trigger:** While in EN_ROUTE, if the technician experiences a breakdown, they click *'Report Hurdle & Cancel'* and select a reason (Vehicle Puncture, Engine Fault, Extreme Waterlogging).
2. **Transaction Isolation:** The system does not cancel the booking. It isolates the failed worker, updates their availability to OFFLINE, and increments their audit log.
3. **Immediate Re-Escalation:** The Spatial Dispatch Service queries PostGIS excluding previously declined workers and immediately dispatches the booking to Worker #2. The citizen is notified: *'Technician A experienced a vehicle breakdown. Reassigned to nearest certified guild member Technician B (2.1 km away).'*

B. The Dynamic On-Site Material Procurement Sequence

1. **In-App Request:** Technician identifies worn-out part, opens the material drawer, inputs item name (e.g., *Havells 32A Contactor*), estimated price (■450), and snaps a photo of the burnt part.
2. **Citizen Pre-Authorization:** The citizen's app receives an interactive push prompt displaying the photo and price. Citizen clicks Approve & Escrow Pre-Authorize.
3. **Cooperative Pass Issuance:** The technician's app generates a **Digital Procurement QR**. The technician visits an authorized partner cooperative hardware merchant, scans the pass, and receives the part.
4. **Settlement:** The partner store POS transmits an API webhook with the digital GST receipt. Citizen pays zero retail markup, and the platform retains a 4% wholesale referral margin.

C. Emergency SOS & Police Telemetry Sequence

1. **One-Touch Activation:** Citizen presses the red SOS button. The app compiles an encrypted incident payload containing: live GPS coordinates, technician's Aadhaar hash, PACS registration number, vehicle plate, and active audio buffer.
2. **Dual Automated Escalation:** The backend simultaneously initiates: (a) An automated telephony bridge to the 24/7 Seva Setu Command Center; (b) A secure webhook transmission to the Police Control Room (PCR 112 API); (c) SMS emergency broadcast to pre-registered family contacts.
3. **Session Lock:** The technician's app is locked into an audit inspection state, preventing session termination until cleared by the control room.

6. RESTful API Contracts & Endpoint Specifications

Sample Core Route Payloads:

```
// 1. POST /api/v1/bookings (Create Booking & Request Quote)
REQUEST: {
  "service_code": "ELEC_REPAIR", "sub_service_name": "MCB_FUSE_REPLACEMENT",
  "urgency_tier": "P1_CRITICAL", "pricing_horizon": "HOURLY", "payment_mode": "UPI_ESCROW",
  "customer_latitude": 28.6315, "customer_longitude": 77.2167, "is_ngo_booking": false
}
RESPONSE (201 Created): {
  "booking_id": "9b1deb4d-3b7d-4bad-9bdd-2b0d7b3dcb6d",
  "status": "BROADCASTING", "start_otp": "4982", "end_otp": "7134",
  "transparent_invoice": {
    "base_labor": 240.00, "toolkit_fee": 60.00, "transit_km": 2.4, "transit_tariff": 30.00,
    "emergency_surcharge": 120.00, "gst_tax": 12.00, "platform_tech_fee": 8.40,
    "federation_cess": 3.60, "total_payable": 474.00, "worker_guaranteed_takehome": 390.00
  }
}

// 2. POST /api/v1/bookings/{id}/verify-start (Doorstep Arrival Handshake)
REQUEST:  { "entered_otp": "4982", "worker_lat": 28.6316, "worker_lng": 77.2166 }
RESPONSE: { "success": true, "new_status": "IN_PROGRESS", "geofence_verified": true }
```

7. Core Algorithmic Logic & Mathematical Formulations

A. Geospatial Proximity & Triage Ranking Function

Candidate workers within radius R are scored using a weighted multi-variable ranking function:

```
def calculate_worker_rank(worker: CooperativeWorker, booking: Booking) -> float:
    dist_km = haversine(booking.pickup_coords, worker.current_location)
    proximity_score = 100.0 / (dist_km + 1.0)
    reliability_weight = (1.0 - worker.rejection_rate_14d) * 50.0
    rating_weight = (worker.rating / 5.0) * 30.0
    triage_bonus = 0.0
    if booking.urgency_tier == UrgencyTier.P1_CRITICAL:
        if worker.is_emergency_certified and worker.rejection_rate_14d < 0.10:
            triage_bonus = 150.0
        else:
            triage_bonus = -500.0  # Exclude uncertified/unreliable workers from critical calls
    return proximity_score + reliability_weight + rating_weight + triage_bonus
```

B. The Dynamic Subsidy Allocation Guardrail Algorithm

Prevents unbacked platform subsidies by cross-checking real-time district reserve balances and monthly institutional quotas:

```
def evaluate_subsidy_entitlement(citizen: Citizen, bill_amount: float, district_pool_bal: float) -> tuple:
    if not citizen.is_ngo_verified:
        return 0.0, 'STANDARD_RATE'
    if citizen.used_subsidy_hours_curr_month >= citizen.monthly_subsidy_quota_hours:
        return 0.0, 'QUOTA_EXHAUSTED_STANDARD_RATE'
    subsidy_target = bill_amount * 0.20
    if district_pool_bal >= subsidy_target:
        return subsidy_target, 'SUBSIDY_APPROVED_FROM_COMMERCIAL_POOL'
    else:
        return 0.0, 'QUEUED_FOR_OFF_PEAK_CSR_SPONSORSHIP'
```

8. Concurrency, Security & Anti-Fraud Specifications

Security / Concurrency Vector	Implementation Architecture	Anti-Fraud Guarantee & Protection Mechanism
Worker Dispatch Race Conditions	Redis Redlock Distributed Locking with 5-second TTL	Guarantees atomic worker claim; prevents two workers from accepting the same booking.
State Transition Integrity	PostgreSQL Optimistic Row Versioning (OCC)	Rejects stale state transitions; throws 409 Conflict if row version mismatches.
False Photo Proof of Work	EXIF Hardware Geolocation & Timestamp Diff	Refuses END_OTP if photo coordinates deviate >250m from pickup site or <20 mins elapsed.
OTP Theft / Brute Force	SHA-256 Cryptographic Hashing with Salt	Zero plaintext OTPs in database; 3 invalid attempts triggers 10-minute cooldown.
Aadhaar Privacy Compliance	UIDAI Zero-Knowledge Tokenization via Digilocker	No raw 12-digit Aadhaar stored; only one-way cryptographic tokens persisted.

Low-Level Design Certification: This LLD specification fulfills all technical, operational, and algorithmic requirements for Seva Setu. The architecture is fully decoupled, horizontally scalable across PostgreSQL PostGIS spatial nodes, resilient to en-route real-world network failures via fallback state machines, and mathematically guaranteed against unbacked platform subsidies.